

CHPC SCC26 Project Overview

Note: GitHub repositories serve as the central source of truth for the project.

End goal:

Build a Collab Agent that acts as a conversational interface on Discord and Telegram to securely spawn and despawn Kubernetes agents executing authenticated tasks.

Duration

The project runs for six weeks, with the **first four weeks focusing on standard infrastructure deployment across all teams**, followed by two weeks of project-specific development.

Node Architecture

The cluster design will **resemble something of this ilk:**

- **Edge Node:** 4 vCPU, 10 GB RAM, 50 GB Storage. Hosts WireGuard, Pi-hole/DNS, nftables, Wazuh manager, and Suricata.
- **API Load Balancer:** 2 vCPU, 4 GB RAM, 25 GB Storage. Hosts HAProxy as a stable Kubernetes API endpoint.
- **K8s Control Plane (x1):** 4 vCPU, 8 GB RAM, 30 GB Storage.
- **K8s Workers (x2):** 8 vCPUs, 16 GB RAM, 40 GB storage each for platform, workbench, and project workloads.

Team Roles

Each member must research and be responsible for one of the following domains to start: It is important all members understand eachothers content and roles.

Proposed Roles for members are:

- Kei:

- **Infrastructure Deployment:** OpenStack, Terraform, and networking design (firewalls, security groups, DNS).

- Kieran

- **Cloud Automation:** Ansible, Kubernetes, Cilium, and Cinder.

- Pride

- **CI/CD, Telemetry, and Security:** ArgoCD, observability via Prometheus and Grafana, and security via Wazuh and Suricata.

- Erin

- **Front-end and AI Agents:** Astro web frontend, Agent Control Plane (ACP), and project specialization.

Timeline

WEEK	FOCUS	END GOALS
1	OpenStack, Terraform, Ansible	Infrastructure fundamentals and an automated, repeatable environment deployment
2	Kubernetes substrate and GitOps	Kubernetes running on a control plane and two workers, with Cilium, Cinder volumes, Argo CD, Sealed Secrets, and DNS/TLS through Traefik
3	Observability, telemetry, security, Astro	Prometheus and Grafana dashboards, Wazuh and Suricata running, and a basic website we can log in to
4	Agent Control Plane	A self-hosted Hermes Agent, reachable from the Astro frontend, that can spawn Hermes workers on Kubernetes and call models on the A100 and H200 nodes

WEEK	FOCUS	END GOALS
5	Collab Agent track	Discord chat adapter, authenticated ACP task, and result preservation across agent deletion and respawn
6	Report and reproducibility	A two-page technical journal article, a project poster, and a full teardown and respawn of the infrastructure as proof
7	Consolidation	Finishing, fixing, and polishing

More on the roles

DevOps Infrastructure

Terraform and Ansible are used to automate the deployment of nodes on OpenStack. Our cluster design depends on our project and our own research, so the final architecture is ours to decide. A Slurm setup may also be part of the cluster, depending on what our research shows.

Cloud Automation

Ansible, Kubernetes, Cilium (Kubernetes networking), and Cinder (OpenStack block storage used for persistent Kubernetes volumes). This area takes the machines from the infrastructure layer and turns them into a working Kubernetes cluster, including building and deploying our own container images.

CI/CD, Telemetry and Security

Argo CD for continuous delivery, secure GitHub-based deployment, Prometheus and Grafana for observability, and Wazuh and Suricata for security monitoring. This area is what gives us visibility into what the system is doing and protects it while it does it.

Frontend and AI Agents

The Astro web frontend, the Agent Control Plane (ACP), the Hermes agent workers, and the Discord and Telegram integrations. For our track this is where the project-specific work happens.
